Qian et al. 2020 · Breast Cancer

Cancer-cell embryonic-program manifold analysis (CCCA six-step protocol)

GO / NO-GO GATES

Gate 1 — Batch mixing & manifold continuity — GO

All 7 patients traverse one continuous manifold: the shared developmental axis (signature PC1, 54% var) is spanned by 7/7 patients (minor samples alone cover 78%); module variance from patient identity only R^2=0.05–0.15. iLISI is imbalance-limited by the 65%-dominant tumour (ceiling 2.26 of 7); the call is led by the embedding + biology, not a scalar.

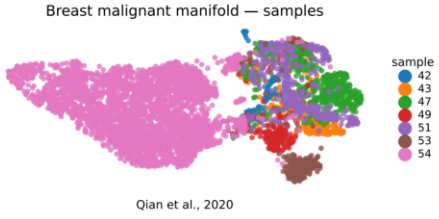
Gate 2 — Embryonic module geography — GO (non-branching)

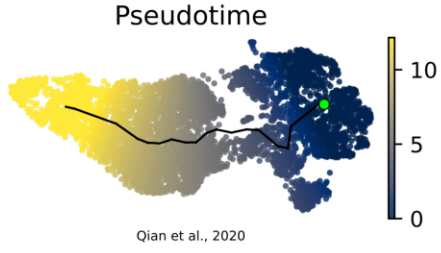
The embryonic programs form one developmental geography: a spatial Gast↔Crest inverse gradient (smoothed r = −0.49, batch-confirmed within patients), Gast & pSom overlapping toward the origin. An honest PPT tree put every tip at negative Gast — no Gast arm, no Gast↔pSom hub. Crest is an inverse gradient, not a Gast-arm enrichment: no inversion flag.

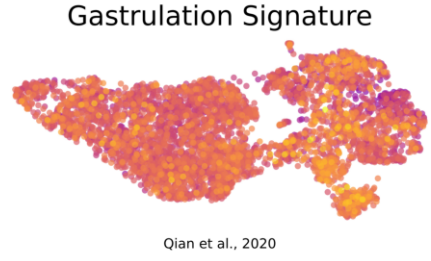
Gate 3 — Hub Gast↔pSom cross-repression — NA

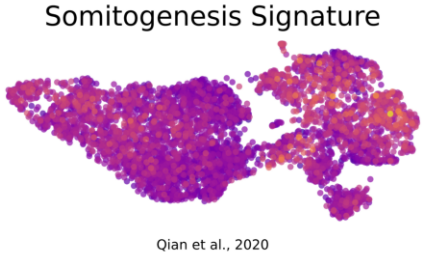
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply.

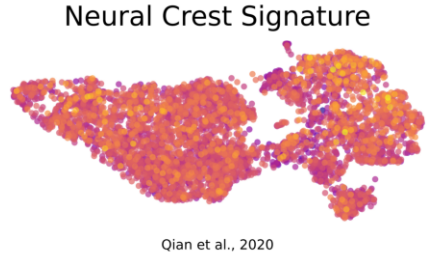
STEP FIGURES

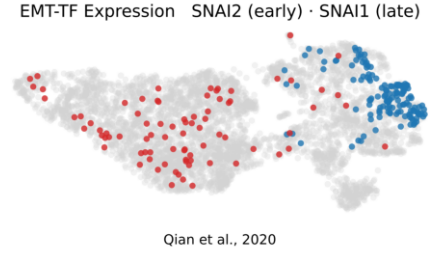












Dataset. Breast cancer, 10x (Qian et al. 2020); 4,801 malignant cells, 7 patients (treatment-naive).

Verdict. The embryonic programs reproduce as a NON-BRANCHING manifold: gastrulation and somitogenesis overlap toward a developmental origin, with neural crest forming a spatial inverse gradient to gastrulation (Crest↔Gast r = −0.49) and no Gast/pSom hub or bifurcation. Geometry = Linear-pSom/Crest. Status: Gate 1 GO · Gate 2 GO · Gate 3 NA. EMT-TF axis SNAI2 (early/origin) → SNAI1 (late), detection-gated panel.

Parameter choices

Malignant cells (raw):
4,818 → 4,801 (thresholded, de-islanded)

Sample threshold:
≥100 cells (gap rule mis-fires: dominant high-side cliff) → 7/7

Overdispersed genes:
1,225 (scFates find_overdispersed)

Integration:
Harmony on sample, θ=10, n_pcs=20 (iLISI-sweep derived)

Embedding:
UMAP spread 1.3, n_neighbors 30, seed 1

Island removal:
radius-graph 6× median-NN → 17 cells / 1 speck

Trajectory:
ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.99)

Root:
Gast/pSom-high, Crest-low tip (developmental origin; Crest late)

Geometry:
Linear-pSom/Crest (non-branching)

EMT-TFs shown:
SNAI2 (origin) · SNAI1 (late), detection-gated panel